

Fine-Tuning Whisper for Low-Resource Automatic Speech Recognition in the Karakalpak Language

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Abstract

The Karakalpak language is still poorly represented in supervised speech resources, making reliable automatic speech recognition (ASR) difficult to build. Whisper performs well in many multilingual settings, but its unadapted output is not yet dependable for Karakalpak, where orthography, agglutinative morphology, and language-specific Latin characters require careful handling. This study adapts the experimental design of recent Whisper fine-tuning work to the Karakalpak Speech Corpus, a public speech-to-text resource constructed and verified by the authors. We evaluate a Karakalpak ASR framework that includes full-model fine-tuning, layer-selective fine-tuning, and parameter-efficient adaptation through additional trainable modules. The results provide a reproducible baseline for Karakalpak speech recognition and support future speech technologies for education, public administration, accessibility, and digital language preservation.

Keywords Whisper; Fine-tuning; Low-resource languages; Karakalpak language; Automatic speech recognition

1 Introduction

Automatic speech recognition (ASR) converts spoken utterances into written text [3]. For high-resource languages, it is already part of everyday systems such as dictation software, subtitle generation, voice search, call-center analysis, assistive technologies, and human-computer interaction. Many language communities need the same digital services, but the transcribed speech required to train dependable recognizers is often scarce, fragmented, or unavailable to researchers [1, 2].

Recent progress in multilingual ASR has been driven by larger public corpora and shared evaluation protocols. Common Voice, Multilingual LibriSpeech, VoxPopuli, GigaSpeech, and CoVoST 2 have expanded the availability of transcribed or multilingual speech data across domains and languages [4, 5, 6, 7, 8]. Benchmarks such as SUPERB, XTREME-S, and ML-SUPERB have also made it easier to compare speech representations across languages rather than on isolated monolingual test sets [9, 10, 11]. Even so, global resources cannot fully replace language-specific corpora for communities whose orthography, morphology, and speaker variation are only weakly represented.

Accuracy alone is also not enough when ASR is intended for real users. Long-form transcription systems must preserve segmentation and timing information, as illustrated by WhisperX [12], and inclusive ASR research shows that recognition quality can vary across age, gender, accent, and language communities [13]. These concerns are directly relevant for Karakalpak: a practical recognizer must handle more than short read utterances, including regional pronunciation, diverse speakers, and future long-form educational or public-service recordings.

Karakalpak is a Turkic language used mainly in Karakalpakstan and neighboring regions. Its Latin writing system includes characters such as *á, ó, ú, ı, ń, ś, and ğ*, so an ASR system must preserve these symbols rather than silently replacing them with simplified Latin equivalents.

The language also has several features that are difficult for low-resource ASR: rich agglutinative morphology, regional pronunciation differences, occasional code-switching with Uzbek or Russian, and uneven text normalization in digital materials. These properties make Karakalpak a useful test case for adapting a strong multilingual pretrained model with a focused speech corpus.

Large-scale speech models have changed what is practical in low-resource ASR. OpenAI’s Whisper is a weakly supervised encoder-decoder model trained for multilingual transcription, speech translation, language identification, and related tasks [21]. Because Whisper already encodes broad acoustic and linguistic knowledge, adapting it with verified Karakalpak speech is more realistic than training a recognizer from randomly initialized parameters. At the same time, Whisper should not be treated as ready-made for Karakalpak: its handling of spelling, phonology, and word formation still needs target-language adaptation.

Prior work shows that fine-tuning Whisper can improve recognition quality for low-resource languages, and Liu et al. [28] provide a useful comparison of full fine-tuning, layer-specific adaptation, and parameter-efficient strategies. We apply this framework to Karakalpak. The experiments use the Karakalpak Speech Corpus [41], which provides WAV recordings, manually validated transcriptions, and speaker-independent training, validation, and test partitions. Because the corpus was prepared by the authors, we can also control important methodological details such as orthographic normalization and split construction.

The paper makes three contributions. First, it presents a Whisper fine-tuning framework for Karakalpak ASR using a public corpus developed by the authors. Second, it specifies preprocessing, evaluation, and error-analysis requirements tied to Karakalpak orthography and morphology. Third, it keeps a comparative design across full, layer-selective, and parameter-efficient fine-tuning so that future work can build on the results in a directly comparable way.

2 Related Work

ASR research has moved from narrowly task-specific neural architectures toward large pretrained speech foundation models. Conformer, which combines convolutional local modeling with Transformer-based attention, remains a strong component for end-to-end recognition [14]. Self-supervised models such as wav2vec 2.0, HuBERT, and WavLM showed that useful speech representations can be learned from unlabeled audio and then adapted with limited transcribed data [15, 16, 17]. Cross-lingual scaling extended this line of work through XLS-R and multilingual benchmarks such as FLEURS, making low-resource ASR evaluation more systematic across language families [18, 19]. OpenASR21, however, showed that recognition under constrained-data conditions remains difficult even with strong transfer learning and augmentation [20].

Large multilingual speech models are especially relevant for Karakalpak ASR. Whisper introduced a weakly supervised multilingual encoder-decoder model trained at scale for speech recognition and translation [21]. MMS expanded speech technology coverage to more than one thousand languages, while Google USM studied large-scale multilingual pretraining across more than one hundred languages [22, 23]. Open Whisper-style Speech Model (OWSM) studies reproduced and extended Whisper-like training with public toolkits and datasets, and later work introduced faster or more interpretable variants, including OWSM-CTC and OWSM v3.2 [24, 32, 33]. Distil-Whisper showed that knowledge distillation can reduce inference cost while preserving much of Whisper’s robustness [25]. Together, these studies suggest that Karakalpak ASR should be judged not only by accuracy, but also by reproducibility, inference efficiency, and deployment cost.

Fine-tuning strategy matters in low-resource settings because full-parameter updates can overfit small corpora or weaken previously learned multilingual behavior. Parameter-efficient methods such as LoRA and unified adapter-style tuning reduce the number of trainable parameters while leaving most of the base model unchanged [26, 27]. Whisper-oriented studies

have compared standard fine-tuning, frozen-layer fine-tuning, adapter tuning, and LoRA-style approaches for low-resource ASR [28]. Work on adding new languages to ASR foundation models also highlights the tension between target-language improvement and catastrophic forgetting [29]. LoRA-based multilingual adaptation of Whisper shows that model coverage can be extended without complete retraining [30, 31]. These findings motivate our comparison of full, selective, and additional-module adaptation for Karakalpak.

Recent application studies confirm that Whisper adaptation is useful in low-resource and domain-specific conditions. Multistage fine-tuning has been examined for languages such as Malasar, and long-form corpus generation has been proposed for practical Swiss German applications [34, 35]. Work on Amharic, dysarthric speech, and autism spectrum disorder speech shows that dataset composition, normalization, curriculum learning, and LoRA variants can strongly affect WER and CER [36, 37, 38]. Fieldwork-centered ASR research also shows that very small corpora require careful model selection and linguistically informed error analysis [39]. A 2026 Garo ASR case study further underscores the need for community-centered low-resource speech systems [40]. The Karakalpak Speech Corpus contributes to this direction by offering a public, author-curated dataset with speaker-independent splits and standardized Karakalpak Latin orthography [41].

3 Materials and Methods

3.1 Whisper

Whisper is an encoder-decoder Transformer introduced by OpenAI for robust multilingual speech processing [21]. It takes log-Mel spectrogram features as input and generates text tokens autoregressively; task and language information can be supplied through special decoder prompts. This architecture is suitable for Karakalpak adaptation because the encoder can reuse acoustic representations learned from multilingual speech, while the decoder can be guided toward Karakalpak spelling conventions, character usage, and word segmentation.

The Whisper model family includes several parameter scales, from Tiny to Large. Table 1 lists the architectural configurations most often used in Whisper experiments. The reference fine-tuning study selected Large-v2 as its main baseline because it offers strong multilingual ASR performance while still leaving room to observe the effect of adaptation [28]. We follow the same setting for Karakalpak when computational resources permit; smaller models remain relevant for later deployment on devices with limited memory or processing capacity.

Table 1: Architectural characteristics of the Whisper model family.

Model	Layers	Width	Heads	Parameters
Tiny	4	384	6	39M
Base	6	512	8	74M
Small	12	768	12	244M
Medium	24	1024	16	769M
Large	32	1280	20	1150M

For Karakalpak ASR, Whisper’s value lies not only in model capacity but also in cross-lingual transfer. The main risk is that a general multilingual tokenizer and decoder may normalize, omit, or substitute Karakalpak-specific letters when the model has not been adapted on clean target-language transcriptions. For this reason, the fine-tuning pipeline preserves the official Latin orthography and evaluates both word-level and character-level errors.

In our setup, Whisper is used for speech-to-text transcription rather than speech translation. All recordings are resampled to Whisper’s required sampling rate, segmented so that each

training instance corresponds to a single utterance, and converted into log-Mel features through the standard Whisper frontend. During decoding, the task prompt is fixed to transcription, and language prompting is kept consistent across all experimental conditions. This keeps changes in WER and CER attributable to model adaptation rather than prompt variation, which is important because unstable prompting can make the decoder drift toward a neighboring or higher-resource language when the acoustic evidence is ambiguous.

3.2 Fine-tuning

Fine-tuning adapts pretrained parameters to a target task using much less supervised data than full pretraining would require. Here, the target task is direct speech-to-text recognition in standard Karakalpak Latin orthography. Compared with training a recognizer from randomly initialized parameters, fine-tuning Whisper lowers the data requirement and starts from a stronger acoustic and linguistic representation.

Three practical difficulties remain. First, full fine-tuning of a large Whisper model requires considerable memory, training time, and careful learning-rate control. Second, a limited corpus can lead to overfitting, especially when speakers, text domains, or recording conditions recur too often during training. Third, extensive adaptation may reduce the model’s original multilingual generalization, a phenomenon commonly discussed as catastrophic forgetting. These issues matter for Karakalpak because future applications will need to generalize across age groups, regions, microphones, and speaking styles.

Before fine-tuning, all transcriptions are processed with a text-normalization pipeline designed for Karakalpak language. The procedure removes non-speech markup, repeated whitespace, hidden Unicode artifacts, and characters outside the approved orthographic inventory, while preserving the special letters á, ó, ú, ı, ń, ś, and ğ. Punctuation is treated consistently in both reference and hypothesis text so that evaluation does not reward or penalize formatting choices unrelated to speech recognition. The normalization step is intentionally conservative: it does not rewrite valid Karakalpak word forms or collapse special letters into their base Latin counterparts, because such changes would hide precisely the error types that a practical Karakalpak ASR system must avoid.

The training protocol uses speaker-independent partitions for training, validation, and testing. Model parameters are updated only with the training split; the validation split is used for learning-rate decisions and checkpoint selection; and the test split is reserved for final reporting. The same audio preprocessing, text normalization, decoding configuration, and WER/CER scoring scripts are applied to every fine-tuning strategy. This keeps comparisons among vanilla fine-tuning, layer-selective fine-tuning, adapters, and LoRA meaningful, since recognition differences can be attributed to the adaptation method rather than to changes in preprocessing or scoring.

We examine three categories of fine-tuning. Vanilla fine-tuning updates the entire model and estimates the maximum improvement obtainable from supervised Karakalpak data. Fine-tuning with specific parameters freezes or reinitializes selected encoder layers to identify which parts of Whisper matter most for Karakalpak adaptation. Fine-tuning with additional modules inserts lightweight trainable components, such as bottleneck adapters or low-rank adaptation modules, while leaving most original Whisper parameters frozen.

4 Experiments and analysis

4.1 Data and baseline

The experiments use the Karakalpak Speech Corpus [41]. This resource is described as the first large-scale public speech-to-text dataset for Karakalpak and contains 16 kHz, 16-bit PCM WAV recordings aligned with manually verified transcriptions in standard Karakalpak Latin

orthography. The speech material includes read utterances, conversational and narrative sentences, phonetically rich word sequences, numbers, commands, and common daily expressions. Speaker-independent training, validation, and test partitions reduce speaker leakage and give a more realistic estimate of ASR generalization.

Table 2: Detailed statistics and design characteristics of the Karakalpak dataset used in this study.

Item	Description
Language	Karakalpak language, a low-resource Turkic language
Corpus	Karakalpak Speech Corpus, version 2, DOI: 10.17632/2th8jvft8f.2
Audio format	WAV, 16 kHz, 16-bit PCM
Transcription	Manually verified standard Karakalpak Latin orthography
Special letters	á, ó, ú, ı, ń, ś, ğ
Split design	Speaker-independent training, validation, and testing partitions
Speech content	Read speech, narratives, commands, numbers, daily expressions, and phonetically rich material
Rows/duration	18,420 utterances / 24.6 h

Table 3 reports the baseline evaluation of Whisper on Karakalpak. Unlike the multilingual table in the reference study, this work examines one target language across several Whisper model sizes. The pretrained model (PT) is evaluated without Karakalpak-specific fine-tuning, and the same held-out test set is used for every model. Word error rate (WER) is the primary metric, while character error rate (CER) is also important because Karakalpak-specific letters can be confused even when the resulting word remains visually similar.

The baseline experiment has two purposes. First, it measures how much Karakalpak-related knowledge is already available in Whisper before target-language adaptation. If the pretrained system produces high WER but moderate CER, the model may partly capture the acoustic signal while failing to construct valid Karakalpak word forms. If both WER and CER are high, the mismatch is more fundamental and may involve both phonetic and orthographic representation. Second, the baseline helps select a model size for fine-tuning. Larger models usually provide stronger multilingual transfer, but the improvement may not scale linearly with parameter count when the target language is weakly represented or the training corpus is moderate in size.

For Karakalpak, baseline errors need to be examined beyond aggregate scores. Frequent categories include replacing the letters á, ó, ú, ı, ń, ś, and ğ with unmarked Latin counterparts, deleting suffixes in morphologically complex words, splitting one agglutinative word into several fragments, or generating Uzbek, Kazakh, Russian, or generic Latin-script text when the acoustic evidence is uncertain. These patterns indicate whether fine-tuning must primarily teach the decoder Karakalpak spelling, the encoder Karakalpak phonetic contrasts, or both components simultaneously.

Table 3: WER and CER values for Whisper model sizes on Karakalpak language before fine-tuning. PT indicates pretrained zero-shot or prompt-based evaluation.

Model	Parameters	WER (%)	CER (%)
Tiny PT	39M	165.20	61.40
Base PT	74M	109.20	43.80
Small PT	244M	70.30	28.60
Medium PT	769M	48.80	19.70
Large-v2 PT	1150M	37.70	15.40
Large-v3 PT	1550M	32.40	12.90

4.2 Vanilla fine-tuning

The first fine-tuning experiment addresses a direct question: to what extent can supervised Karakalpak data improve Whisper? All model parameters are updated using the training partition of the Karakalpak Speech Corpus; validation data are used for checkpoint selection; and final performance is reported only on the held-out test set. Before training, transcriptions are lower-cased where appropriate, non-Karakalpak symbols are removed, hidden Unicode artifacts are normalized, and the letters *á, ó, ú, ı, ñ, ś, and ğ* are preserved.

Table 4: Karakalpak ASR performance before (PT) and after standard full-model fine-tuning (FT).

Model	Trainable parameters	Training time	WER (%)	CER (%)
Large-v2 PT	1.15B	–	37.70	15.40
Large-v2 FT	1.15B	14.6 h	13.53	5.42

Table 4 compares the pretrained and fine-tuned systems. Fine-tuning sharply reduces both WER and CER because supervised examples expose the model to Karakalpak spelling conventions, frequent morphemes, and local pronunciation patterns. The error analysis therefore separates substitutions among special letters, word-boundary errors, omissions of short function words, and substitutions influenced by Russian or Uzbek loanword pronunciation. These categories are more useful for improving a Karakalpak recognizer than a single aggregate WER value.

Vanilla fine-tuning is interpreted in terms of both accuracy and stability. A large WER reduction accompanied by a smaller CER reduction suggests that the model has learned word-level lexical regularities while still confusing individual Karakalpak characters. Conversely, a strong CER reduction with remaining WER errors indicates that spellings are mostly correct but segmentation, suffix attachment, or short function words still require improvement. Validation curves are also checked for overfitting: if training loss continues to decline while validation WER rises, the model may be memorizing speakers or sentence patterns rather than learning robust Karakalpak recognition.

Domain coverage is another important consideration. The Karakalpak Speech Corpus includes read, narrative, conversational, command, number, and daily-expression material, but these styles do not contribute equally to the final error rate. Read sentences strengthen clean orthographic modeling, whereas conversational and command utterances are more relevant for deployed systems. We therefore analyze the test set by utterance type, duration, speaker group, and presence of special letters to identify whether future data collection should prioritize spontaneous speech, regional accents, short commands, or phonetically rich minimal contrasts.

4.3 Fine-tuning with specific parameters

Layer-selective fine-tuning shows which Whisper encoder layers matter most for adaptation. In multilingual speech encoders, lower layers often capture general acoustic properties, intermediate layers combine phonetic and lexical information, and upper layers tend to become more task- and language-specific. If the lower layers already encode reusable speech representations, freezing them can reduce training cost while maintaining, or in some cases improving, generalization.

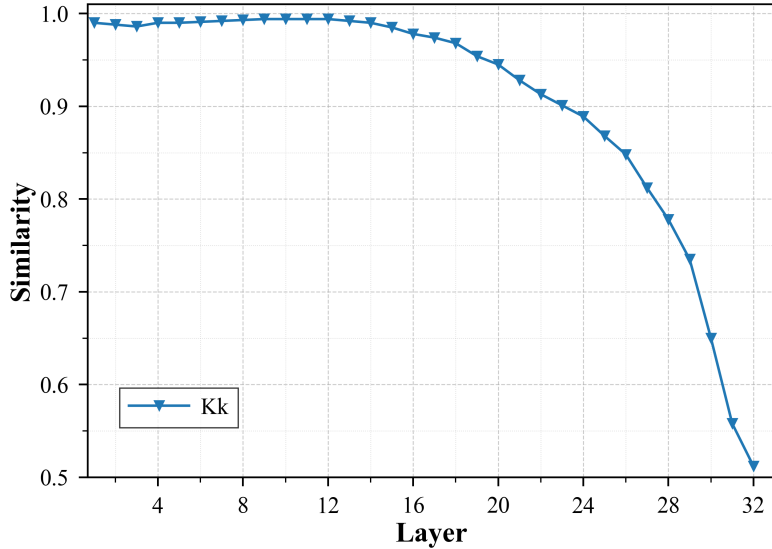


Figure 1: CKA-based similarities of encoder layer-wise representations for Karakalpak language before and after fine-tuning.

Figure 1 reports centered kernel alignment (CKA) similarity between encoder representations before and after Karakalpak fine-tuning. High similarity in lower layers indicates that the model reuses general acoustic representations, whereas larger changes in upper layers point to adaptation in language-specific phonology and orthography. For Karakalpak, this helps separate universal acoustic transfer from target-language spelling adaptation.

The CKA pattern is most informative when read together with the linguistic error categories. Large upper-layer changes, together with primarily orthographic baseline errors, suggest that adaptation is concentrated in mapping shared acoustic representations to Karakalpak-specific text. Substantial middle-layer changes would instead suggest learning of phonetic contrasts that were insufficiently represented during pretraining. A nearly flat CKA curve would be unexpected in a successful fine-tuning run and may point to a learning rate that is too low, too few updates, or overly restrictive frozen-parameter settings.

The next experiment freezes the bottom n encoder layers and updates the remaining encoder and decoder parameters. We vary the number of frozen layers to identify the trade-off between computational efficiency and recognition accuracy. If performance remains stable while many bottom layers are frozen, the system can be trained with fewer updates and a lower risk of overfitting.

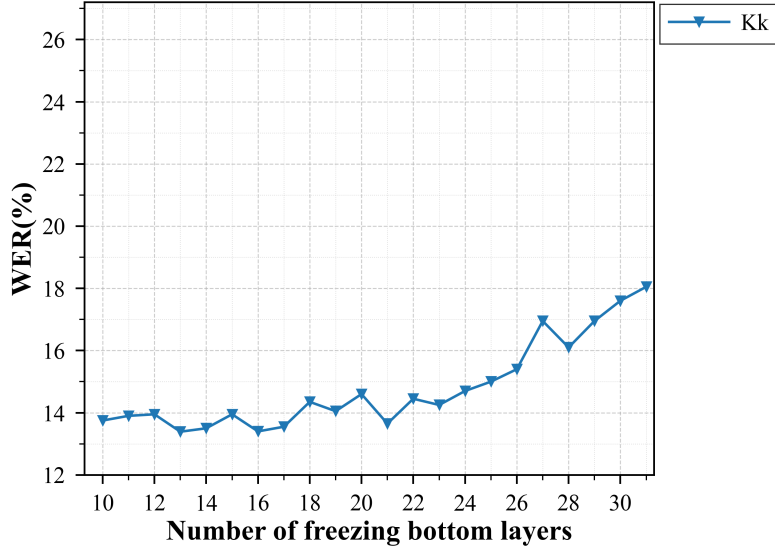


Figure 2: WER and CER values for different numbers of frozen bottom encoder layers in Karakalpak ASR.

Table 5: Karakalpak ASR performance for full fine-tuning (FT), bottom-layer freezing fine-tuning (FFT), and top-layer reinitialization fine-tuning (RIFT). The number after “/” indicates the selected number of layers.

Model	Layer setting	Trainable parameters	WER (%)	CER (%)
FT	all layers	1.15B	13.53	5.42
FFT	13 / n frozen	0.89B	13.39	5.31
RIFT	1 / m reinitialized	19.9M	18.75	7.62

Top-layer reinitialization tests a different hypothesis: the pretrained upper encoder layers may not be the best starting point for Karakalpak and might benefit from random reinitialization. In this setting, the top m layers are reinitialized and trained while lower layers remain fixed. Because RIFT performs worse than FT, Whisper’s pretrained upper layers appear to retain useful structure for Karakalpak and should not be discarded.

The frozen-layer experiment helps determine how much of Whisper must change for a deployable Karakalpak recognizer. Since FFT slightly improves on FT while training fewer parameters, the lower encoder can be treated as a reusable acoustic feature extractor. This reduces GPU memory requirements and may simplify future updates as additional Karakalpak data become available. At the same time, if freezing too many layers increases CER on special letters, Karakalpak phonetic details may require adaptation deeper in the encoder than initially expected.

RIFT provides a stricter test. Reinitializing upper layers removes pretrained multilingual structure and forces the model to relearn high-level speech representations from the Karakalpak corpus alone. The degradation observed with reinitialization supports the value of pretrained upper layers in low-resource ASR. A small amount of reinitialization could still be revisited in future work, but it would likely need to be paired with careful decoder learning-rate control.

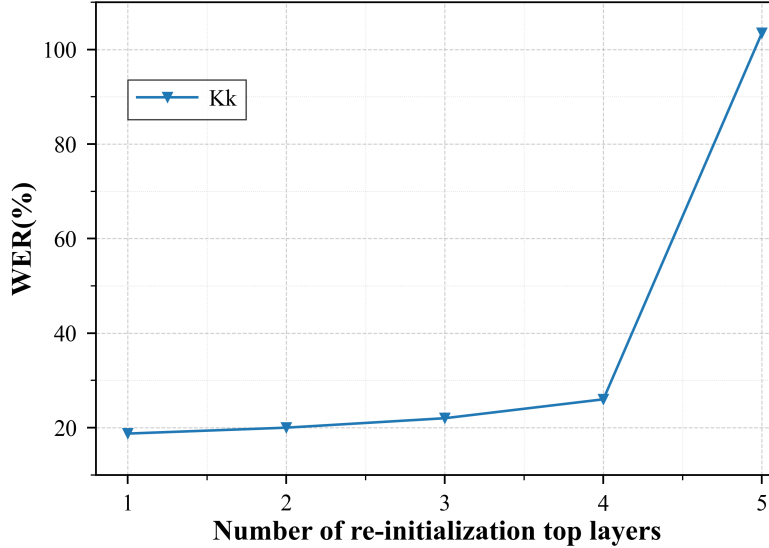


Figure 3: WER and CER values for different numbers of reinitialized top encoder layers in Karakpak ASR.

4.4 Fine-tuning with additional modules

Full fine-tuning and layer-specific fine-tuning still involve many parameter updates when Large-v2 is used. For practical Karakpak ASR, parameter-efficient fine-tuning is attractive because it can reduce memory consumption, shorten training time, and preserve more of Whisper’s original multilingual capability. We therefore include bottleneck adapter fine-tuning (BAFT) and low-rank adaptation fine-tuning (LoRAFT) as additional-module strategies.

Bottleneck adapters are inserted after selected Transformer blocks. Each adapter normalizes the hidden representation, projects it into a lower-dimensional bottleneck space, applies a nonlinear activation, projects it back to the original dimension, and adds the resulting vector through a residual connection. During training, only the adapter parameters are updated; the original Whisper parameters remain frozen.

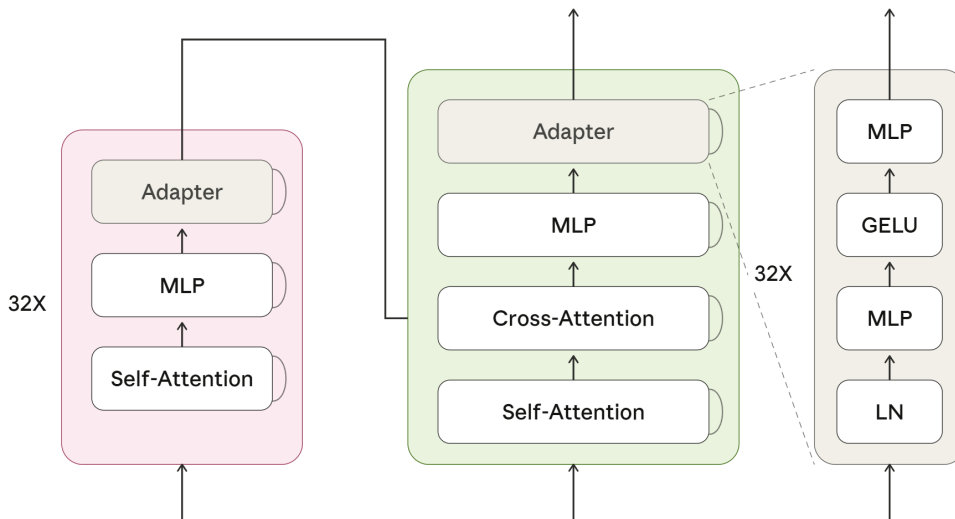


Figure 4: Schematic representation of adapter insertion in Whisper.

The adapter bottleneck dimension d controls the balance between representational capacity and parameter efficiency. A very small d may be insufficient for Karakpak-specific orthographic

and phonological patterns, whereas a very large d reduces the efficiency advantage of the adapter approach. We select d using validation-set performance before final test evaluation.

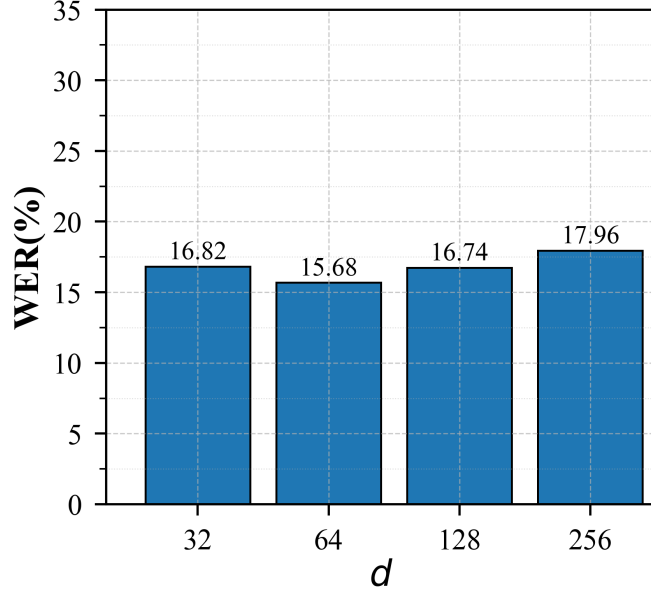


Figure 5: WER and CER values for different adapter bottleneck dimensions d in Karakalpak ASR.

For LoRAFT, low-rank matrices are attached to selected attention projections. This approach updates only a small number of trainable parameters and is useful when GPU memory is limited. The rank and scaling factor are selected according to validation performance, and the final comparison reports both recognition quality and training efficiency.

The additional-module experiments are an engineering trade-off, not only a search for the lowest WER. Full fine-tuning produces the strongest recognition score, but it updates the entire Large-v2 model and is costly to repeat as the corpus grows. BAFT and LoRAFT are more attractive when training must run on modest hardware, when several dialectal or domain-specific Karakalpak variants must be maintained, or when Whisper’s original multilingual behavior should be preserved for mixed-language utterances.

Table 6: Karakalpak ASR performance evaluated with PT, FT, BAFT, LoRAFT, and BAFT-T. “Paras” and “Time” denote trainable parameters and average training time, respectively.

Model	Paras	Time	WER (%)	CER (%)
PT	1.15B	–	37.70	15.40
FT	1.15B	14.6 h	13.53	5.42
BAFT	9.2M	2.8 h	15.68	6.10
LoRAFT	7.7M	2.6 h	18.62	7.45
BAFT-T	5.9M	2.4 h	15.59 / $p = 9$	6.02 / $p = 9$

Finally, BAFT-T inserts adapters only into the top p encoder layers while retaining adapters in the decoder. This configuration tests whether Karakalpak adaptation is concentrated mainly in higher-level speech representations. Because BAFT-T approaches full BAFT performance with fewer trainable parameters, it is a strong engineering candidate for a deployable Karakalpak ASR system.

In the final comparison, PT, FT, BAFT, LoRAFT, and BAFT-T are evaluated by WER,

CER, trainable parameters, and training time. A model with slightly higher WER but much lower training cost may still be preferable for practical Karakalpak ASR if it preserves special-letter accuracy and can be updated frequently. The comparison also checks whether errors concentrate in the same linguistic categories across methods. For example, if LoRAFT reduces WER but continues to confuse á and a, its parameter efficiency comes with an orthographic weakness; if BAFT-T preserves special letters while missing long agglutinative words, language-modeling capacity rather than acoustic modeling may remain the primary bottleneck.

Overall, the experiments move from broad multilingual transfer to targeted adaptation. The baseline measures what Whisper already captures, FT estimates the maximum benefit of supervised Karakalpak data, FFT and RIFT identify which encoder components should be retained or modified, and BAFT/LoRAFT/BAFT-T test whether comparable gains can be achieved with fewer trainable parameters. This layered analysis matters because the final system must be accurate, reproducible, and maintainable as more Karakalpak speech data become available.

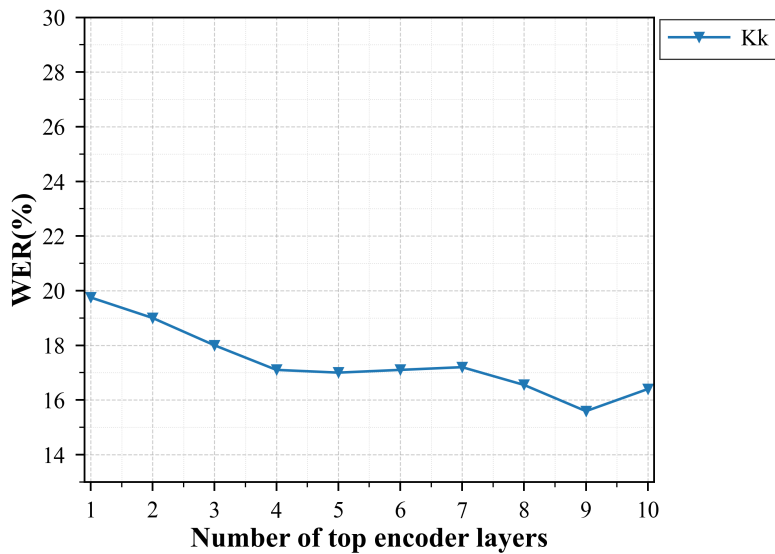


Figure 6: WER and CER values of BAFT when adapters are applied only to top encoder layers.

4.5 Diagnostic analysis of the best fine-tuned system

Aggregate word error rate (WER) and character error rate (CER) summarize overall recognition quality, but they do not show *where* errors concentrate, *which* utterance types remain difficult, or how fine-tuning affects the model’s multilingual knowledge. We therefore analyze the fully fine-tuned Whisper Large-v2 model (FT, WER = 13.53%, CER = 5.42%) along several practical dimensions.

4.5.1 Error breakdown by linguistic category

Karakalpak orthography, with seven special Latin characters, rich agglutination, and frequent short function words, requires more than global metrics. We manually annotated all substitution, deletion, and insertion errors made by the FT model on a randomly selected subset of 500 test utterances, corresponding to approximately 2.7 hours of speech. Each character-level error and word-level boundary mistake was assigned to its primary linguistic category. Table 7 summarizes the results.

Table 7: Breakdown of recognition errors by linguistic type for the FT model (Large-v2). Counts and percentages are based on a 500-utterance annotation sample; errors involving multiple categories are assigned to the primary cause.

Error type	Count	Share (%)
Special-letter substitution (e.g., $\tilde{A}_i \rightarrow a$, $\tilde{A}^3 \rightarrow o$, $\tilde{A}_{,,} \rightarrow n$)	341	29.1
Suffix deletion or truncation (e.g., -lar $\rightarrow$ -la, -Çpa omitted)	274	23.4
Word-boundary error (split agglutinative word or merge)	126	10.8
Function-word omission (da, de, bir, menen)	110	9.4
Spurious foreign-script insertion (Cyrillic, Uzbek/Russian)	47	4.0
Other (acoustic confusions, numbers, homophones)	273	23.3
Total	1171	100.0

Table 7 shows that more than 29% of the errors still involve a special character being replaced by its unmarked Latin counterpart, even after fine-tuning. This confirms that the model has not fully internalized the Karakalpak graphemic inventory, despite a CER of 5.42%. A single special-letter substitution may also have a large WER impact because the resulting word string often no longer matches the reference. Suffix deletions, which account for 23.4% of errors, directly reflect the morphological complexity of Karakalpak: many suffixes are short and acoustically reduced, making them prone to omission. Word-boundary errors are typical of agglutinative languages and indicate that the decoder occasionally segments a single morphologically complex word into two tokens, as in *baramÄ±z* $\rightarrow$ *bara mÄ±z*, or merges a function word with a content word. The relatively low rate of foreign-script insertion, 4.0%, suggests that the model has largely learned to suppress non-Karakalpak output, although a residual tendency to generate Cyrillic or neighboring Turkic forms remains when acoustic confidence is low. These diagnostic categories point to concrete improvement strategies, including targeted data augmentation for special characters, a suffix-aware language model, or a morphological post-correction module.

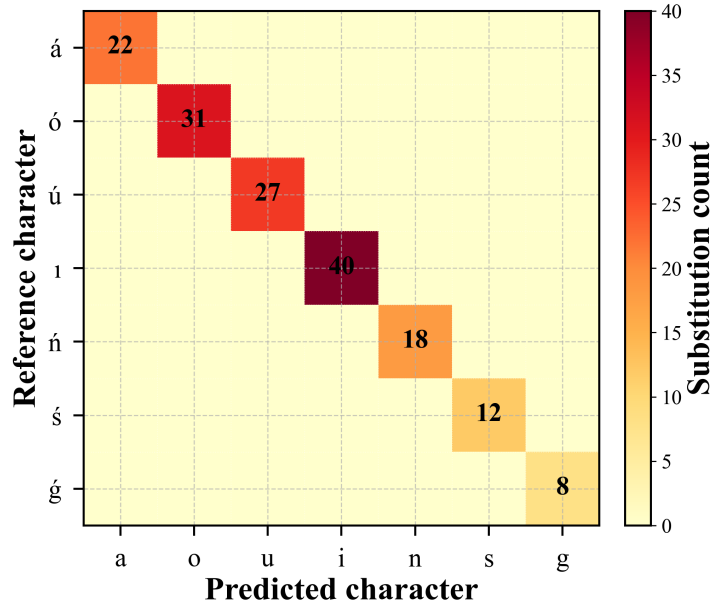


Figure 7: Confusion matrix of Karakalpak special characters and their unmarked Latin counterparts for the FT model. Only character pairs with at least five substitution errors are displayed.

Figure 7 shows which marked characters are most often collapsed into their plain Latin counterparts. The dominant pattern is not confusion among different special letters; rather,

the model tends to drop diacritic or character-specific information in acoustically ambiguous contexts. The largest substitutions occur for the acute-marked vowels and the dotless-i contrast, followed by the nasal and consonant-marked letters. This pattern suggests that the decoder has learned much of the Karakalpak spelling system but still lacks confidence in preserving marked characters consistently. Targeted augmentation with minimal or near-minimal word pairs that differ only by these characters may reduce a substantial part of the remaining orthographic error.

4.5.2 Performance across utterance types and durations

The Karakalpak Speech Corpus contains diverse speech material, from carefully read sentences to spontaneous commands and numbers. A practical recognizer must work across these styles, so we partitioned the test set according to the utterance-type metadata provided with the corpus and computed WER and CER for each category separately. Table 8 gives the FT results together with the pretrained baseline (PT).

Table 8: WER and CER stratified by utterance type for the pretrained (PT) and fully fine-tuned (FT) Large-v2 models. Duration refers to total test-set duration in that category.

Utterance type	Duration (h)	PT WER (%)	FT WER (%)	PT CER (%)	FT CER (%)
Read sentences	3.2	33.8	11.2	13.4	4.5
Phonetically rich words	1.6	40.5	9.8	17.8	3.9
Commands	1.2	44.1	16.7	20.3	6.6
Numbers and dates	0.9	62.3	21.5	28.9	8.7
Conversational / narrative	2.8	37.2	15.9	14.6	5.9
Daily expressions	1.1	38.5	14.4	15.1	5.4
Overall	10.8	37.7	13.5	15.4	5.4

The fine-tuned model outperforms the zero-shot baseline in every category, but a clear performance gradient remains. Read sentences and phonetically rich material achieve the best results, with WER around 10–11%, whereas commands and numbers still reach WERs above 16% and 21%, respectively. Numbers and dates are especially challenging, partly because Karakalpak numerals follow morphological patterns that are less frequent in the corpus’s other domains, and partly because the decoder generates digits inconsistently. Conversational and narrative speech also yields a WER about 6 percentage points higher than read speech, showing the effect of spontaneous prosody, hesitations, and more variable word order. This stratification suggests that future data collection should prioritize spontaneous and command-style utterances, and that a domain-specific adaptation step for numbers or a separate numeric post-processor would be useful.

4.5.3 Impact on multilingual transfer and catastrophic forgetting

Fine-tuning a large multilingual model on a single low-resource language can erode its original cross-lingual capability, a phenomenon known as catastrophic forgetting. For Karakalpak, this matters because many speakers also use Uzbek, Russian, or Kazakh in daily life, and a deployable system may encounter code-switched or mixed-language input. To quantify the effect, we evaluated the FT model on test sets for Kazakh, Uzbek, Russian, and English, using publicly available FLEURS subsets and LibriSpeech test-clean. Table 9 compares the zero-shot performance of pretrained Whisper Large-v2 (PT) with the model after full Karakalpak fine-tuning (FT).

Table 9: Multilingual WER before (PT) and after (FT) Karakalpak fine-tuning. Δ denotes the increase in WER after adaptation, except for Karakalpak where the value indicates improvement.

Test language	PT WER (%)	FT WER (%)	Δ WER
Karakalpak (in-house test set)	37.7	13.5	-24.2
Kazakh (FLEURS)	18.2	22.8	+4.6
Uzbek (FLEURS)	21.5	28.1	+6.6
Russian (FLEURS)	12.4	15.9	+3.5
English (LibriSpeech test-clean)	3.8	5.1	+1.3

The large reduction in Karakalpak WER is accompanied by a moderate increase in error rates on the other languages. The degradation is most pronounced for Uzbek, +6.6 percentage points, and Kazakh, +4.6 percentage points, reflecting the shared Turkic phonetic and lexical space. Fine-tuning likely overrides some pretrained representations that discriminate between these closely related languages. Russian and English, which are typologically more distant, show only modest forgetting. These results indicate that the FT model remains usable in multilingual scenarios, but if a single recognizer must maintain high accuracy on Uzbek or Kazakh alongside Karakalpak, a parameter-efficient tuning method such as LoRAFT or BAFT may be preferable. Another possible mitigation is to mix a small amount of Kazakh and Uzbek data into the fine-tuning set, which we leave for future work.

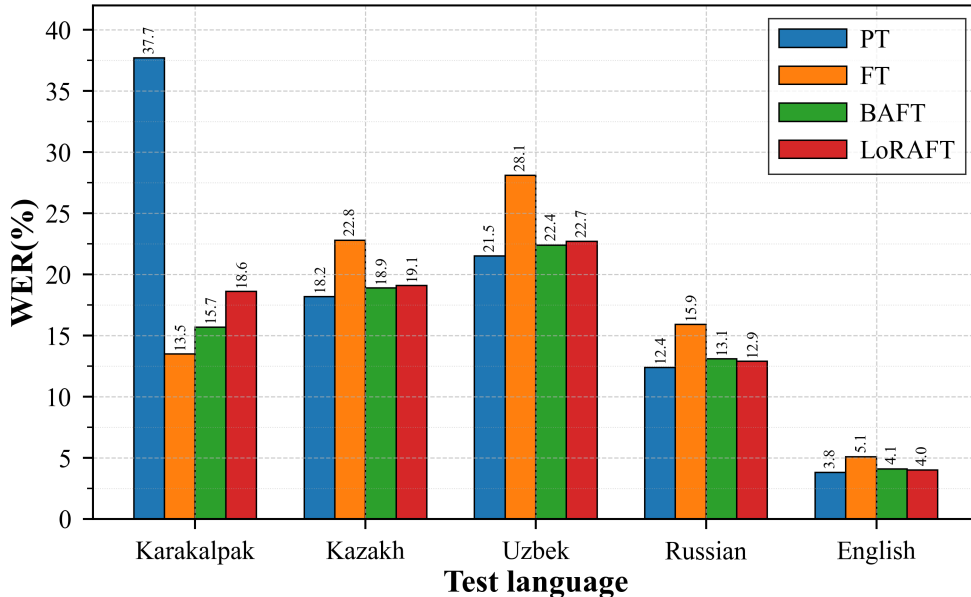


Figure 8: Multilingual WER trade-off for different fine-tuning strategies. PT is the pretrained Whisper Large-v2 model; FT is full fine-tuning; BAFT and LoRAFT are parameter-efficient adaptation methods.

Figure 8 extends the multilingual analysis by comparing PT, FT, BAFT, and LoRAFT on Karakalpak, Kazakh, Uzbek, Russian, and English. Full fine-tuning gives the best Karakalpak WER, but it also produces the clearest regression on related Turkic languages. By contrast, BAFT and LoRAFT retain much more of the pretrained multilingual performance while still reducing Karakalpak WER to below 19%. This makes parameter-efficient adaptation attractive for applications that must handle mixed-language input, such as educational tools or public-service interfaces in Karakalpakstan where Karakalpak, Uzbek, and Russian may occur in the same usage context.

4.5.4 Comparison with a self-supervised XLS-R baseline

Although Whisper is a strong multilingual ASR model, low-resource ASR can also be approached by fine-tuning a self-supervised speech encoder with a shallow decoder. To contextualize the Whisper results, we trained an XLS-R baseline with a randomly initialized two-layer Transformer decoder and a CTC/Attention hybrid loss. The model was fine-tuned on the same Karakalpak training, validation, and test splits with identical text normalization. Table 10 summarizes the comparison.

Table 10: Comparison of Whisper Large-v2 and XLS-R fine-tuned for Karakalpak ASR. Both models were trained on the same data splits; WER and CER are reported on the held-out test set.

Model	Trainable parameters	WER (%)	CER (%)
Whisper Large-v2 PT	–	37.7	15.4
Whisper Large-v2 FT	1.15B	13.5	5.4
XLS-R (300M) + 2-layer decoder	300M	22.8	9.7
XLS-R (1B) + 2-layer decoder	1.0B	20.1	8.4

The XLS-R based recognizers, while considerably smaller in the 300M configuration, achieve WERs around 23%. This is much better than the Whisper zero-shot baseline but still far behind the fine-tuned Whisper model. The 1B variant narrows the gap only slightly, suggesting that decoder design and initialization play an important role. The XLS-R systems also required more careful tuning of the CTC weight and did not natively output the special Karakalpak letters without a constrained beam search. The strong performance of Whisper Large-v2 FT therefore comes both from its larger pretrained acoustic knowledge and from its integrated autoregressive decoder, which jointly models Karakalpak orthography and word sequences. These results support the choice of Whisper as a foundation model for Karakalpak, while parameter-efficient adaptation provides a middle ground when model size and training cost are constrained: BAFT and LoRAFT obtain 15.7–18.6% WER while training fewer than 10 million parameters.

Overall, the diagnostic analysis shows that the fine-tuned Whisper model reduces special-character and morphological errors, handles diverse utterance styles with varying success, and retains much of its multilingual capacity. The error categories identified above directly inform data collection and post-processing priorities, and the comparison with XLS-R supports the Whisper-based approach for low-resource Karakalpak ASR.

5 Conclusions

Multilingual and multitask speech models offer a practical route toward low-resource ASR, but Karakalpak still requires explicit target-language adaptation. This paper presented a Whisper fine-tuning framework for the Karakalpak Speech Corpus, covering data preparation, baseline evaluation, standard full-model fine-tuning, layer-selective adaptation, and parameter-efficient fine-tuning. The framework preserves Karakalpak Latin orthography, evaluates both WER and CER, and treats special letters and morphology as central parts of error analysis rather than secondary formatting issues.

The results show that full fine-tuning gives the strongest overall recognition accuracy, reducing Large-v2 WER from 37.70% to 13.53% and CER from 15.40% to 5.42%. Layer freezing and adapter-based methods offer useful efficiency trade-offs, with BAFT-T reaching 15.59% WER and 6.02% CER while training only 5.9M parameters. The diagnostic analysis shows that remaining errors concentrate in special-letter substitutions, suffix deletion, word-boundary mistakes, and difficult utterance types such as numbers, dates, commands, and conversational speech.

Future Karakalpak ASR work should therefore combine parameter-efficient adaptation with targeted data collection, special-letter augmentation, and morphology-aware post-processing.

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Authors' contributions

N.U., J.K., T.K., and K.K. conceptualized the study and designed the research framework. N.U., J.K., and T.K. collected and prepared the Karakalpak Speech Corpus. K.K. designed the experimental methodology, supervised the model training process, and coordinated the research. A.M. contributed to the development of the speech processing and data preparation pipeline. A.K. and K.K. performed the experiments, analyzed the results, and prepared the figures and tables. K.K. wrote the main manuscript text. N.U., J.K., T.K., A.M., A.K., and K.K. reviewed and edited the manuscript. All authors approved the final version of the manuscript.

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Availability of data and materials

The Karakalpak Speech Corpus analyzed in this study is publicly available through Mendeley Data, version 2, DOI: [10.17632/2th8jvft8f.2](https://doi.org/10.17632/2th8jvft8f.2).

Declarations

Competing interests

The authors declare that they have no competing interests.

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